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ANIMALS

## This spider uses trapped fireflies to lure in more prey

Male fireflies ensnared in the spider's web flash their lanterns like females, luring more males



An orb weaver spider (*Araneus ventricosus*) wraps a male firefly (*Abscondita terminalis*) that got stuck in its web. The spider's venom or its bite may cause the firefly's flashing lights to blink in a femalelike pattern, luring other males into the web.

XINHUA FU

**By Erin Garcia de Jesús**

3 HOURS AGO

Sometimes fireflies shouldn't follow the light.

**SPLASHING. SLIDING.  
SAVINGS.**

A single flash from a female usually helps male *Abscondita terminalis* fireflies find her among tall grasses at dusk. Males are showier, giving off multiple light pulses to attract a mate. If ensnared in an orb weaver spider's web, however, flashy males can become deadly decoys.

Orb weaver spiders seem to [trick male fireflies into blinking](#) in a pattern more like females, researchers report August 19 in *Current Biology*. The femalelike lights lure other males into the web, allowing the arachnids to stock up on food.

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"What a cool behavior," says Ximena Nelson, an animal behaviorist at the University of Canterbury in Christchurch, New Zealand, who was not involved with the study. The new observations add a [unique hunting tactic](#) to spiders' repertoires (SN: 9/18/23).

Behavioral ecologist Daiqin Li of Hubei University in Wuhan, China, and colleagues set up cameras on spiders' webs in nearby farmland to watch what happened to fireflies caught in them. The team placed a firefly directly into the orb weavers' spider webs and either left the spider or removed it from its web. After each trial started, the researchers counted how many additional fireflies



In the webs of *Araneus ventricosus* spiders, the fireflies changed their flashes. But in the webs of other spiders, Li says, they did not.

Male fireflies were more likely to get caught in webs and change their flashing pattern when an *A. ventricosus* orb weaver spider was around compared with when it was absent. Those spiders were also less likely to capture more fireflies when Li and his team painted over the insects' flashing lanterns with black ink, the team observed.

Trapped males that had been bitten and wrapped in a spider's silk also stopped using both their lanterns (males have two) to pulse light. Flashing just a single lantern, the males' signals became weaker, single pulses of light, appearing much closer to a female flashing pattern.

The orb weavers also handled fireflies differently than nonflashing beetle species, Li says. While spiders wrapped other beetles in a thick layer of silk and began to feed almost immediately, fireflies got a light wrapping so their lanterns were still visible and then were stocked away in the web.

As more fireflies get caught, Li says, the spiders "repeat the sequence for each of them and leave them there, just flashing."

The jury is still out on how exactly the signals change, Nelson says. Li suspects the spider's bite or venom has something to do with it. He's now planning to test those ideas and look for orb weavers outside of southeast Asia that use this tactic.

#### CITATIONS

X. Fu et al. [Spider manipulates and exploits firefly's bioluminescent signals](https://doi.org/10.1016/j.cub.2024.07.011). *Current Biology*. Published online August 19, 2024. doi: 10.1016/j.cub.2024.07.011.

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